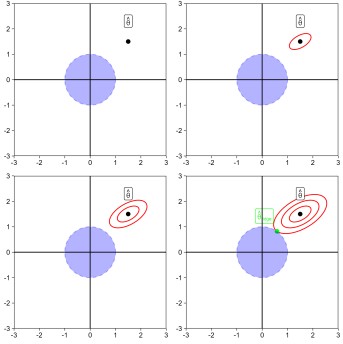
# Introduction to Machine Learning

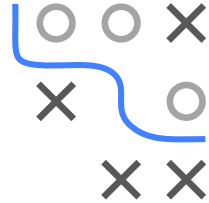
## Regularization
## Ridge Regression




**Learning goals**

- Regularized linear model
- Ridge regression / *L2* penalty
- Understand parameter shrinkage
- Understand correspondence to constrained optimization

# REGULARIZATION IN LM

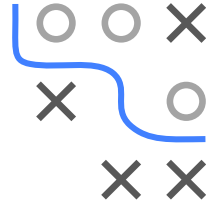


- Can also overfit if $p$ large and $n$ small(er)
- OLS estimator requires full-rank design matrix
- For highly correlated features, OLS becomes sensitive to random errors in response, results in large variance in fit
- We now add a complexity penalty to the loss:

$$\mathcal{R}_{\text{reg}}(\boldsymbol{\theta}) = \sum_{i=1}^{n} \left( y^{(i)} - \boldsymbol{\theta}^\top \mathbf{x}^{(i)} \right)^2 + \lambda \cdot J(\boldsymbol{\theta}).$$

# RIDGE REGRESSION / L2 PENALTY

Intuitive measure of model complexity is deviation from 0-origin; coeffs then have no or a weak effect. So we measure $J(\boldsymbol{\theta})$ through a vector norm, shrinking coeffs closer to 0.



$$
\begin{aligned}
\hat{\boldsymbol{\theta}}_{\text{ridge}} &= \arg\min_{\boldsymbol{\theta}} \sum_{i=1}^{n} \left( y^{(i)} - \boldsymbol{\theta}^T \mathbf{x}^{(i)} \right)^2 + \lambda \sum_{j=1}^{p} \theta_j^2 \\
&= \arg\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|_2^2 + \lambda \|\boldsymbol{\theta}\|_2^2
\end{aligned}
$$

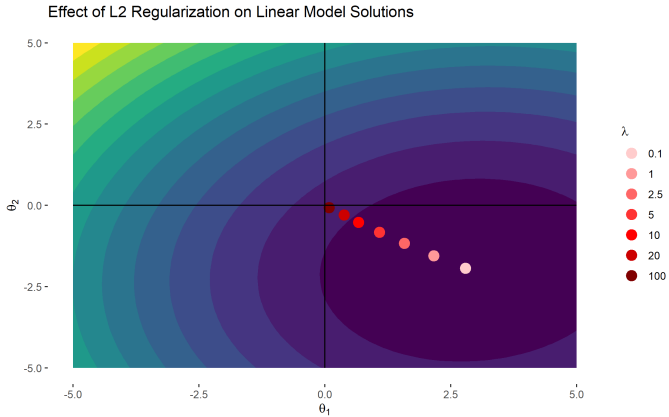
Can still analytically solve this:

$$
\hat{\boldsymbol{\theta}}_{\text{ridge}} = (\mathbf{X}^T\mathbf{X} + \lambda \boldsymbol{I})^{-1} \mathbf{X}^T \mathbf{y}
$$

Name: We add pos. entries along the diagonal "ridge" of $\mathbf{X}^T\mathbf{X}$

# RIDGE REGRESSION / L2 PENALTY / 2

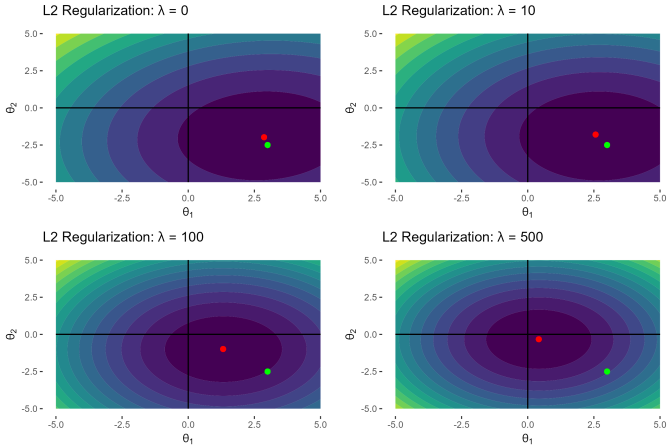
Let $y = 3x_1 - 2x_2 + \epsilon$, $\epsilon \sim N(0, 1)$. The true minimizer is $\theta^* = (3, -2)^T$, with $\hat{\boldsymbol{\theta}}_{\text{ridge}} = \arg\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|^2 + \lambda\|\boldsymbol{\theta}\|^2$.




With increasing regularization, $\hat{\theta}_{ridge}$ is pulled back to the origin (contour lines show unregularized objective).

# RIDGE REGRESSION / L2 PENALTY / 3

Contours of regularized objective for different $\lambda$ values.
$\hat{\boldsymbol{\theta}}_{\text{ridge}} = \arg\min_{\boldsymbol{\theta}} \|\mathbf{y} - \mathbf{X}\boldsymbol{\theta}\|^2 + \lambda\|\boldsymbol{\theta}\|^2$.



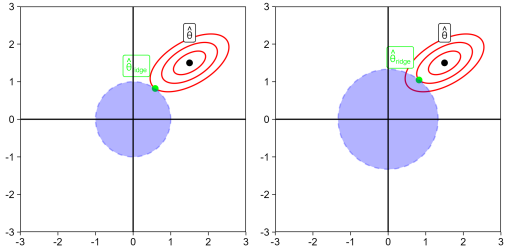

Green = true coefs of the DGP and red = ridge solution.

# RIDGE REGRESSION / L2 PENALTY / 4

We understand the geometry of these 2 mixed components in our regularized risk objective much better, if we formulate the optimization as a constrained problem (see this as Lagrange multipliers in reverse).

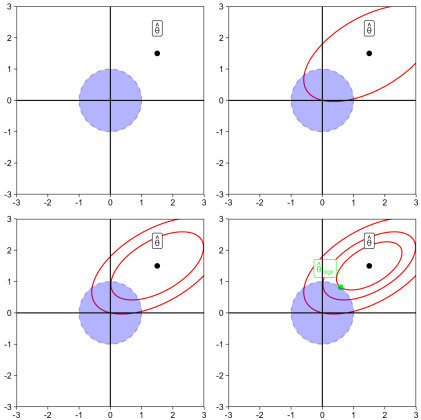
$$\min_{\boldsymbol{\theta}} \quad \sum_{i=1}^{n} \left( y^{(i)} - f\left(\mathbf{x}^{(i)} \mid \boldsymbol{\theta}\right)\right)^2$$

$$\text{s.t.} \quad \|\boldsymbol{\theta}\|_2^2 \leq t$$



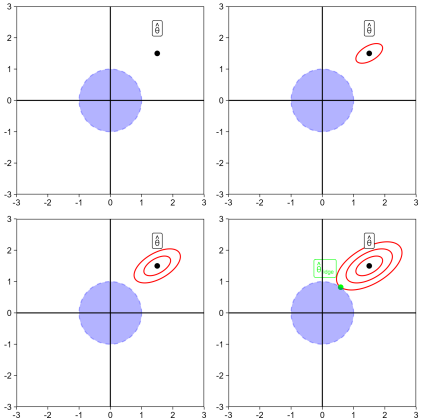

NB: There is a bijective relationship between $\lambda$ and $t$: $\lambda \uparrow \Rightarrow \; t \downarrow$ and vice versa.

# RIDGE REGRESSION / L2 PENALTY / 5



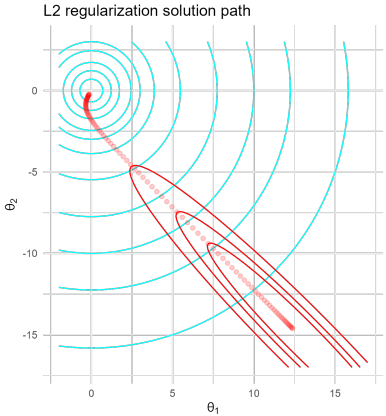

- Inside constraints perspective: From origin, jump from contour line to contour line (better) until you become infeasible, stop before.
- We still optimize the $\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$, but cannot leave a ball around the origin.
- $\mathcal{R}_{\text{emp}}(\boldsymbol{\theta})$ grows monotonically if we move away from $\hat{\boldsymbol{\theta}}$ (elliptic contours).
- Solution path moves from origin to border of feasible region with minimal $L_2$ distance.

# RIDGE REGRESSION / L2 PENALTY / 6




- Outside constraints perspective: From $\hat{\theta}$, jump from contour line to contour line (worse) until you become feasible, stop then.
- So our new optimum will lie on the boundary of that ball.
- Solution path moves from unregularized estimate to feasible region of regularized objective with minimal $L_2$ distance.

# RIDGE REGRESSION / L2 PENALTY / 7




- Here we can see entire solution path for ridge regression
- Cyan contours indicate feasible regions induced by different $\lambda$s
- Red contour lines indicate different levels of the unreg. objective
- Ridge solution (red points) gets pulled toward origin for increasing $\lambda$

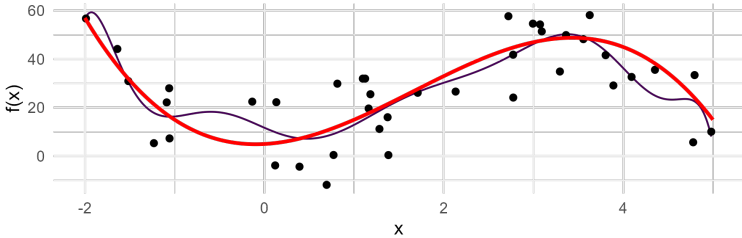
# EXAMPLE: POLYNOMIAL RIDGE REGRESSION

Consider $y = f(x) + \epsilon$ where the true (unknown) function is $f(x) = 5 + 2x + 10x^2 - 2x^3$ (in red).

Let's use a $d$th-order polynomial

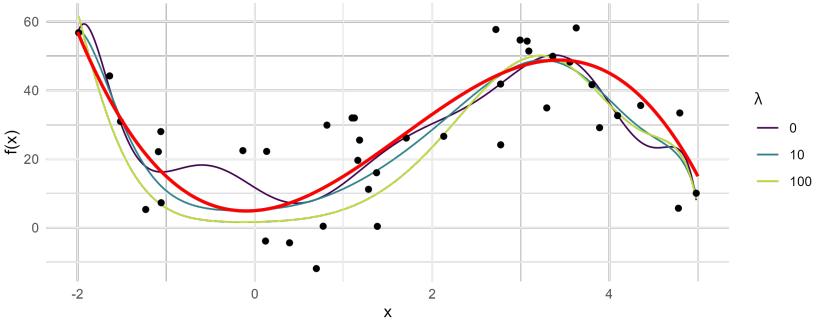
$$f(x) = \theta_0 + \theta_1 x + \cdots + \theta_d x^d = \sum_{j=0}^{d} \theta_j x^j.$$

Using model complexity $d = 10$ overfits:



# EXAMPLE: POLYNOMIAL RIDGE REGRESSION / 2

With an *L2* penalty we can now select *d* "too large" but regularize our model by shrinking its coefficients. Otherwise we have to optimize over the discrete *d*.




| $\lambda$ | $\theta_0$ | $\theta_1$ | $\theta_2$ | $\theta_3$ | $\theta_4$ | $\theta_5$ | $\theta_6$ | $\theta_7$ | $\theta_8$ | $\theta_9$ | $\theta_{10}$ |
|---|---|---|---|---|---|---|---|---|---|---|---|
| 0.00 | 12.00 | -16.00 | 4.80 | 23.00 | -5.40 | -9.30 | 4.20 | 0.53 | -0.63 | 0.13 | -0.01 |
| 10.00 | 5.20 | 1.30 | 3.70 | 0.69 | 1.90 | -2.00 | 0.47 | 0.20 | -0.14 | 0.03 | -0.00 |
| 100.00 | 1.70 | 0.46 | 1.80 | 0.25 | 1.80 | -0.94 | 0.34 | -0.01 | -0.06 | 0.02 | -0.00 |